

# Backend Project Handbook

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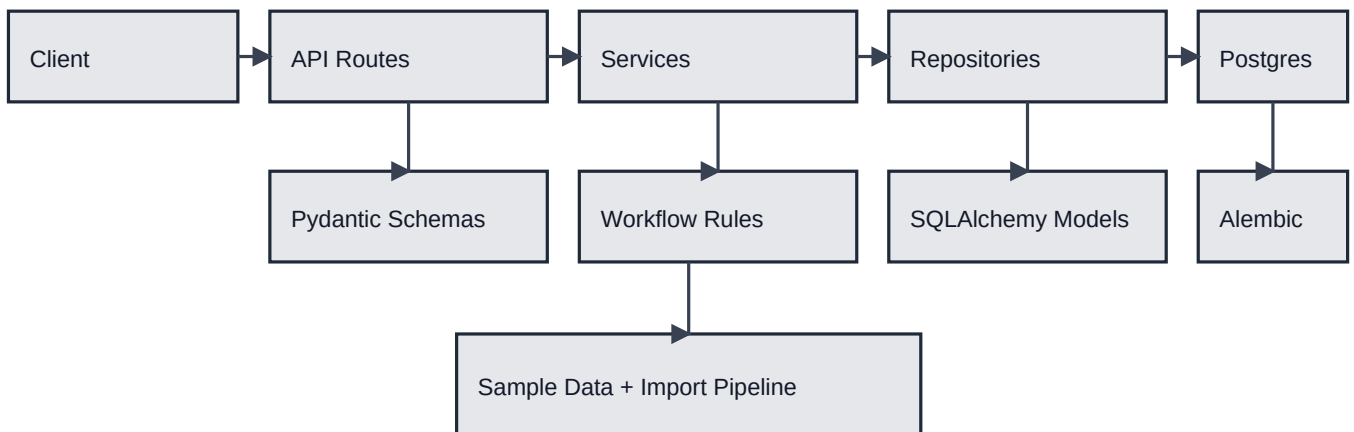
## 1. Executive Summary

This backend is a lead-centric sales intelligence system built on FastAPI, SQLAlchemy, Alembic, and PostgreSQL.

The lead is the central business entity and is connected to customer, campaign, journey, engagement, call, task, follow-up, AI prediction, and decision modules.

## 2. Architecture

### System Architecture



Flow: request in -> domain orchestration -> persistent state -> typed response

- API Layer: app/api/routes
- Service Layer: app/services
- Repository Layer: app/repositories
- Persistence Layer: app/models + PostgreSQL
- Schema Contracts: app/schemas
- DB Evolution: Alembic migrations

Request flow:

Client -> Route -> Service -> Repository -> DB  
-> business rules -> response schema

### Architecture Diagram (Mermaid):

## 3. Feature Modules

1. Health + Dashboard
2. Customers
3. Leads
4. Products
5. Campaigns

6. Engagement Channels
7. Website Journey
8. Calls
9. Tasks + Follow-ups
10. Data Import + Export
11. ML Training + Model Catalog
12. AI Predictions
13. Decision Engine
14. Analytics
15. Reports
16. Lead Lifecycle (assignment, transfer, outcomes, timeline, details)

## **4. Data Processing**

1. Source data is loaded from backend/sample\_data.xlsx using scripts/preload\_sample\_data.py.
2. The loader maps grouped records into products, campaigns, customers, leads, tasks, follow-ups, calls, engagement events, and website events.
3. Live APIs use these persisted records, not synthetic smoke values.
4. AI endpoints score/segment leads and store prediction logs.
5. Decision endpoint combines predictions + rules into next action recommendations.
6. Lifecycle module adds assignment/transfer, outcome tracking, and timeline events to create a connected business workflow.
7. Unified lead details endpoint aggregates lead + customer + campaign + engagement + AI + operational activity for frontend rendering.

### **4.1 Sample Data Columns (sample\_data.xlsx)**

Total columns in dataset sheet: 91

1. Customer\_ID
2. CRM\_Channel
3. CRM\_Department
4. CRM\_Lead\_Type
5. CRM\_Lead\_Source
6. CRM\_Data\_Medium
7. CRM\_Source
8. CRM\_Data\_Source\_Platform
9. CRM\_Product\_Code
10. CRM\_Product\_Name
11. CRM\_UTM\_Source
12. CRM\_UTM\_Medium
13. CRM\_UTM\_Campaign
14. CRM\_Gender
15. CRM\_Age\_Band
16. CRM\_Income\_Band
17. CRM\_Occupation

18. CRM\_Education
19. CRM\_Tobacco\_User
20. CRM\_NonResident\_Flag
21. CRM\_Existing\_Plan\_Flag
22. CRM\_Lead\_Create\_Hour
23. CRM\_Lead\_Create\_DayOfWeek
24. CRM\_Lead\_Create\_Weekend
25. CRM\_Lead\_Create\_Month
26. CRM\_Days\_Create\_To\_Update
27. CRM\_Contact\_Count
28. CRM\_NonContact\_Count
29. CRM\_No\_Of\_Attempts
30. CRM\_Has\_Application\_No
31. CRM\_Has\_Payment\_Flag
32. CDR\_Total\_Calls
33. CDR\_Connected\_Calls
34. CDR\_Connect\_Rate
35. CDR\_Total\_Talk\_Sec
36. CDR\_Avg\_Talk\_Sec
37. CDR\_Max\_Talk\_Sec
38. CDR\_Distinct\_Call\_Days
39. CDR\_Distinct\_Agents
40. CDR\_Callbacks\_Scheduled
41. CDR\_Inbound\_Calls
42. CDR\_Top\_Call\_Status
43. CDR\_Top\_Q\_Type
44. CDR\_First\_Call\_Delay\_Days
45. CDR\_Call\_Span\_Days
46. MSG\_Campaigns\_Targeted
47. MSG\_Sent
48. MSG\_Delivered
49. MSG\_Read
50. MSG\_Clicked
51. MSG\_Replied
52. MSG\_Failed
53. MSG\_Engaged
54. WEB\_Tracked
55. Visits
56. Page\_Views
57. Total\_Seconds\_Spent
58. ev\_Lead\_Creation
59. ev\_Lead\_Submitted
60. ev\_Calculator\_Start
61. ev\_Retrieve\_Quote

62. ev\_Quote\_Proceed  
63. ev\_Personal\_Details\_Next  
64. ev\_Plan\_Details\_Load  
65. ev\_Make\_Payment\_Load  
66. ev\_Pay\_Now  
67. ev\_Payment\_Success  
68. ev\_Payment\_Failure  
69. ev\_Proposal\_Form\_Load  
70. ev\_Proposal\_Form\_Proceed  
71. ev\_Generate\_OTP  
72. ev\_OTP\_Submit  
73. ev\_Brochure\_Download  
74. ev\_Connect\_With\_Expert  
75. ev\_Page\_Scroll\_25  
76. ev\_Page\_Scroll\_95  
77. WEB\_Step\_Name  
78. WEB\_Step\_Number  
79. WEB\_New\_Vs\_Repeat  
80. WEB\_Plan\_Type  
81. WEB\_Plan\_Variant  
82. WEB\_Quoted\_Price  
83. WEB\_Coverage\_Amount  
84. WEB\_Payment\_Frequency  
85. WEB\_Last\_Touch\_Channel  
86. WEB\_UTM\_Source  
87. WEB\_Device\_Type  
88. Label\_Source\_Disposition  
89. Label\_Source\_Lead\_Status  
90. Label\_Basis  
91. LABEL\_Customer\_Validity

## 4.2 Column Meaning (data\_dictionary sheet)

- Customer\_ID | source=generated | Random anonymized row id (mapping kept outside the dataset)
- CRM\_Channel | source=CRM CHANNEL | Acquisition channel of the lead
- CRM\_Department | source=CRM DEPARTMENT | Owning department
- CRM\_Lead\_Type | source=CRM LEAD\_TYPE | Lead type code
- CRM\_Lead\_Source | source=CRM LEAD\_SOURCE | Source system/partner of the lead
- CRM\_Data\_Medium | source=CRM DATA\_MEDIUM | Data medium code
- CRM\_Source | source=CRM SOURCE | Combined source string
- CRM\_Data\_Source\_Platform | source=CRM P\_DATA\_SOURCE | Platform that generated the lead (e.g. PARTNER\_APP)
- CRM\_Product\_Code | source=CRM PROD\_ID | Product id attached to the lead
- CRM\_Product\_Name | source=CRM PRODUCT\_NAME | Product name if recorded
- CRM\_UTM\_Source | source=CRM MISC\_UTM\_SOURCE | Marketing UTM source

- CRM\_UTM\_Medium | source=CRM MISC\_UTM\_MEDIUM | Marketing UTM medium
- CRM\_UTM\_Campaign | source=CRM MISC\_UTM\_CAMPAIGN | Marketing UTM campaign
- CRM\_Gender | source=CRM CUST\_GENDER | Declared gender (demographic, non-identifying)
- CRM\_Age\_Band | source=CRM CUST\_AGE (banded) | Age banded: 18-25/26-35/36-45/46-55/56+/Unknown
- CRM\_Income\_Band | source=CRM ANNUAL\_INCOME (banded) | Annual income banded: <3L/3-5L/5-10L/10-25L/25L+/Unknown
- CRM\_Occupation | source=CRM OCCUPATION | Declared occupation
- CRM\_Education | source=CRM EDUCATION | Declared education
- CRM\_Tobacco\_User | source=CRM TOBACCO\_USER | Tobacco user flag Y/N
- CRM\_NonResident\_Flag | source=CRM NonResident\_FLAG | NonResident flag
- CRM\_Existing\_Plan\_Flag | source=CRM EXISTING\_PLAN\_PRODUCT | Existing plan product if any
- CRM\_Lead\_Create\_Hour | source=CRM LEAD\_CREATE\_DATE (derived) | Hour of day the lead was created (0-23)
- CRM\_Lead\_Create\_DayOfWeek | source=CRM LEAD\_CREATE\_DATE (derived) | Day of week of lead creation
- CRM\_Lead\_Create\_Weekend | source=CRM LEAD\_CREATE\_DATE (derived) | 1 if created on Sat/Sun
- CRM\_Lead\_Create\_Month | source=CRM LEAD\_CREATE\_DATE (derived) | Creation month YYYY-MM
- CRM\_Days\_Create\_To\_Update | source=CRM dates (derived) | Days between lead creation and last update
- CRM\_Contact\_Count | source=CRM CNT\_COUNT | CRM-recorded contact count
- CRM\_NonContact\_Count | source=CRM NCNT\_COUNT | CRM-recorded non-contact count
- CRM\_No\_Of\_Attempts | source=CRM NO\_OF\_ATTEMPTS | CRM-recorded attempt count
- CRM\_Has\_Application\_No | source=CRM APPLICATION\_NO (flag) | 1 if an application number exists (id itself dropped)
- CRM\_Has\_Payment\_Flag | source=CRM COLL\_AMOUNT (flag) | 1 if payment was collected (amount dropped)
- CDR\_Total\_Calls | source=CDR Apr-Jun (agg) | Total dialer call records for this lead
- CDR\_Connected\_Calls | source=CDR (agg) | Calls with duration > 0 (a recording exists for these)
- CDR\_Connect\_Rate | source=CDR (agg) | Connected / total calls
- CDR\_Total\_Talk\_Sec | source=CDR (agg) | Sum of call durations in seconds
- CDR\_Avg\_Talk\_Sec | source=CDR (agg) | Average duration of connected calls
- CDR\_Max\_Talk\_Sec | source=CDR (agg) | Longest single call in seconds
- CDR\_Distinct\_Call\_Days | source=CDR (agg) | Number of distinct days the lead was called
- CDR\_Distinct\_Agents | source=CDR (agg) | Distinct agent count (ids dropped)
- CDR\_Callbacks\_Scheduled | source=CDR Callback Time (agg) | Calls with a callback scheduled
- CDR\_Inbound\_Calls | source=CDR Call Direction (agg) | Inbound call count
- CDR\_Top\_Call\_Status | source=CDR Call Status (agg) | Most frequent call status
- CDR\_Top\_Q\_Type | source=CDR Q Type (agg) | Most frequent queue type
- CDR\_First\_Call\_Delay\_Days | source=CDR + CRM (derived) | Days from lead creation to first call
- CDR\_Call\_Span\_Days | source=CDR (derived) | Days between first and last call
- MSG\_Campaigns\_Targeted | source=camp messaging (agg via customer key) | Distinct messaging campaigns that targeted this customer
- MSG\_Sent | source=camp (agg) | Messages sent
- MSG\_Delivered | source=camp (agg) | Messages delivered
- MSG\_Read | source=camp (agg) | Messages read
- MSG\_Clicked | source=camp (agg) | Link clicks
- MSG\_Replied | source=camp (agg) | Replies
- MSG\_Failed | source=camp (agg) | Failed + bounced messages
- MSG\_Engaged | source=camp (derived) | 1 if any read/click/reply

- WEB\_Tracked | source=WEB var5 (derived) | 1 if this lead had website activity (Apr-May window)
- Visits | source=WEB metrics | Web visits attributed to the lead id
- Page\_Views | source=WEB metrics | Page views
- Total\_Seconds\_Spent | source=WEB metrics | Total seconds on site
- ev\_\* | source=WEB funnel events | Funnel event counts: lead creation/submission, calculator, quote, payment, proposal, OTP, brochure, expert-connect, scroll depth
- WEB\_Step\_Name | source=WEB var31 | Furthest journey step name recorded
- WEB\_Step\_Number | source=WEB var86 | Step number
- WEB\_New\_Vs\_Repeat | source=WEB var3 | New vs repeat visitor
- WEB\_Plan\_Type | source=WEB var35 | Plan type explored
- WEB\_Plan\_Variant | source=WEB var105 | Plan variant explored
- WEB\_Quoted\_Price | source=WEB var90 | Top quoted price
- WEB\_Coverage\_Amount | source=WEB var91 | Top quoted coverage amount
- WEB\_Payment\_Frequency | source=WEB var97 | Payment frequency chosen
- WEB\_Last\_Touch\_Channel | source=WEB lasttouchchannel | Last marketing touch channel
- WEB\_UTM\_Source | source=WEB campaign.utm-source | Web UTM source
- WEB\_Device\_Type | source=WEB mobiledevicetype | Device type used
- Label\_Source\_Disposition | source=CRM DISPOSITION | Raw disposition the label was derived from (drop before training to avoid leakage)
- Label\_Source\_Lead\_Status | source=CRM LEAD\_STATUS | Raw lead status (drop before training to avoid leakage)
- Label\_Basis | source=derived | Which rule produced the label: conversion fields / disposition / lead\_status
- LABEL\_Customer\_Validity | source=derived | TARGET: 0=junk/unreachable, 1=contactable not interested/eligible, 2=interested, 3=converted

### 4.3 Column To API Field Mapping

- Customer\_ID -> customers.external\_customer\_id -> returned in /api/customers and /api/leads/{id}/details.customer
- CRM\_Product\_Code/CRM\_Product\_Name -> products.code/products.name -> returned in /api/products and details.campaign linkage
- CRM\_UTM\_Campaign/CRM\_Data\_Source\_Platform -> campaigns.code/name -> returned in /api/campaigns
- CRM\_Channel/CRM\_Data\_Medium/CRM\_Source -> leads.source\_channel/source\_medium/current\_handler
- Label\_Source\_Lead\_Status/Label\_Source\_Disposition -> leads.status/recommended\_action
- LABEL\_Customer\_Validity/CDR\_Connect\_Rate/Page\_Views/MSG\_Engaged -> leads.lead\_score
- CRM\_Department -> leads.current\_section and lifecycle assignment.to\_section
- CDR\_\* metrics -> calls and engagement event payloads
- WEB\_\* metrics -> website\_events fields and payload
- Label\_Basis/Disposition -> task descriptions + followup notes + timeline details

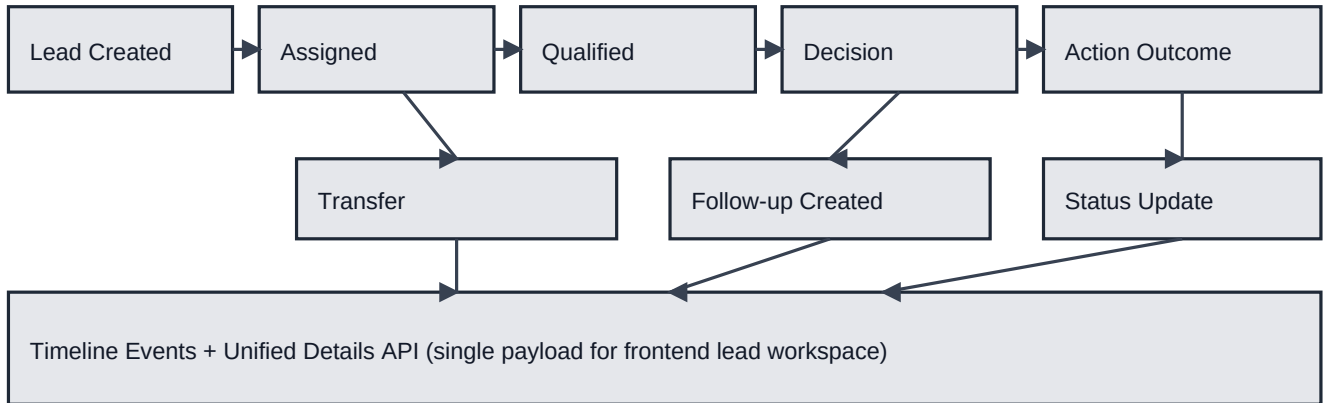
## 5. How To Run In Detail

1. Create backend-local virtual environment and install requirements.
2. Configure .env with DATABASE\_URL for PostgreSQL.
3. Run migrations: ./venv/bin/alembic upgrade head
4. Preload real dataset: ./venv/bin/python scripts/preload\_sample\_data.py --reset
5. Start API server: ./venv/bin/uvicorn app.main:app --host 0.0.0.0 --port 8000
6. Open Swagger: /api/docs and OpenAPI: /api/openapi.json

7. Run smoke verification: `./venv/bin/python scripts/live_api_smoke.py`
8. Run tests: `./venv/bin/pytest -q`
9. Lifecycle-only test: `./venv/bin/pytest -q tests/test_lead_lifecycle.py`

## 6. Feature Workflows And Handling

### Lead Lifecycle Workflow



Outcomes can move lead stage, set recommendation, and trigger new operational tasks.

1. Lead intake: create lead -> initial timeline event -> assignment queue.
2. Assignment/transfer: updates current owner and section; previous assignment marked non-current.
3. AI scoring: prediction APIs persist score outputs per prediction type.
4. Decisioning: next action recommendation combines prediction signal and business logic.
5. Outcome handling: captures result, updates lead stage/status, optionally creates follow-up, and appends timeline events.
6. Reporting: analytics/reports endpoints query aggregated module-level data for operational monitoring.

### Lifecycle Diagram (Mermaid):

## 7. API Inventory And Response Contracts

Total API paths in OpenAPI: 62

### POST /api/ai/analyze-lead

- Summary: Analyze Lead
- Tags: ai
- Request: body=context, customer\_id, lead\_id
- Responses: 200: customer\_id, generated\_at, lead\_id, predictions | 422: detail

### POST /api/ai/conversion

- Summary: Predict Conversion
- Tags: ai
- Request: body=context, customer\_id, lead\_id

- Responses: 200: details, label, model\_name, prediction\_type, rationale, score | 422: detail

### **GET /api/ai/customers/{customer\_id}**

- Summary: Get Customer Predictions
- Tags: ai
- Request: params=customer\_id(path)
- Responses: 200: items[created\_at], items[customer\_id], items[details], items[id], items[label], items[lead\_id], items[model\_name], items[prediction\_type], ... | 422: detail

### **POST /api/ai/intent**

- Summary: Predict Intent
- Tags: ai
- Request: body=context, customer\_id, lead\_id
- Responses: 200: details, label, model\_name, prediction\_type, rationale, score | 422: detail

### **POST /api/ai/lead-score**

- Summary: Predict Lead Score
- Tags: ai
- Request: body=context, customer\_id, lead\_id
- Responses: 200: details, label, model\_name, prediction\_type, rationale, score | 422: detail

### **GET /api/ai/leads/{lead\_id}**

- Summary: Get Lead Predictions
- Tags: ai
- Request: params=lead\_id(path)
- Responses: 200: items[created\_at], items[customer\_id], items[details], items[id], items[label], items[lead\_id], items[model\_name], items[prediction\_type], ... | 422: detail

### **POST /api/ai/next-best-action**

- Summary: Predict Next Action
- Tags: ai
- Request: body=context, customer\_id, lead\_id
- Responses: 200: details, label, model\_name, prediction\_type, rationale, score | 422: detail

### **POST /api/ai/product-recommendation**

- Summary: Predict Product
- Tags: ai
- Request: body=context, customer\_id, lead\_id
- Responses: 200: details, label, model\_name, prediction\_type, rationale, score | 422: detail

### **POST /api/ai/segment**

- Summary: Predict Segment
- Tags: ai



- Request: body=context, customer\_id, lead\_id
- Responses: 200: details, label, model\_name, prediction\_type, rationale, score | 422: detail

## **POST /api/ai/validity**

- Summary: Predict Validity
- Tags: ai
- Request: body=context, customer\_id, lead\_id
- Responses: 200: details, label, model\_name, prediction\_type, rationale, score | 422: detail

## **GET /api/analytics/channels**

- Summary: Get Channels
- Tags: analytics
- Request: none
- Responses: 200: items

## **GET /api/analytics/funnel**

- Summary: Get Funnel
- Tags: analytics
- Request: none
- Responses: 200: conversion\_rate, converted\_leads, lost\_leads, new\_leads, qualified\_leads, total\_leads

## **GET /api/analytics/overview**

- Summary: Get Overview
- Tags: analytics
- Request: none
- Responses: 200: conversion\_rate, converted\_leads, open\_tasks, pending\_followups, total\_calls, total\_campaigns, total\_customers, total\_leads

## **GET /api/calls**

- Summary: List Calls
- Tags: calls
- Request: params=skip(query), limit(query), lead\_id(query), customer\_id(query), status\_filter(query)
- Responses: 200: items[CDR\_Avg\_Talk\_Sec], items[CDR\_Call\_Direction], items[CDR\_Top\_Call\_Status], items[Excel\_Fields], items[created\_at], items[customer\_id], items[ended\_at], items[id], ... | 422: detail

## **POST /api/calls**

- Summary: Create Call
- Tags: calls
- Request: body=CDR\_Call\_Direction, CDR\_Top\_Call\_Status, customer\_id, lead\_id, notes, phone\_number, scheduled\_at
- Responses: 201: CDR\_Avg\_Talk\_Sec, CDR\_Call\_Direction, CDR\_Top\_Call\_Status, Excel\_Fields, created\_at, customer\_id, ended\_at, id, ... | 422: detail

## **GET /api/calls/{call\_id}**

- Summary: Get Call
- Tags: calls
- Request: params=call\_id(path)
- Responses: 200: CDR\_Avg\_Talk\_Sec, CDR\_Call\_Direction, CDR\_Top\_Call\_Status, Excel\_Fields, created\_at, customer\_id, ended\_at, id, ... | 422: detail

### **PATCH /api/calls/{call\_id}**

- Summary: Update Call
- Tags: calls
- Request: params=call\_id(path) ; body=customer\_id, direction, lead\_id, notes, phone\_number, scheduled\_at, status
- Responses: 200: CDR\_Avg\_Talk\_Sec, CDR\_Call\_Direction, CDR\_Top\_Call\_Status, Excel\_Fields, created\_at, customer\_id, ended\_at, id, ... | 422: detail

### **DELETE /api/calls/{call\_id}**

- Summary: Delete Call
- Tags: calls
- Request: params=call\_id(path)
- Responses: 204: Successful Response | 422: detail

### **POST /api/calls/{call\_id}/end**

- Summary: End Call
- Tags: calls
- Request: params=call\_id(path) ; body=ended\_at, notes
- Responses: 200: CDR\_Avg\_Talk\_Sec, CDR\_Call\_Direction, CDR\_Top\_Call\_Status, Excel\_Fields, created\_at, customer\_id, ended\_at, id, ... | 422: detail

### **POST /api/calls/{call\_id}/start**

- Summary: Start Call
- Tags: calls
- Request: params=call\_id(path) ; body=started\_at
- Responses: 200: CDR\_Avg\_Talk\_Sec, CDR\_Call\_Direction, CDR\_Top\_Call\_Status, Excel\_Fields, created\_at, customer\_id, ended\_at, id, ... | 422: detail

### **GET /api/campaigns**

- Summary: List Campaigns
- Tags: campaigns
- Request: params=skip(query), limit(query)
- Responses: 200: items[CRM\_UTM\_Campaign], items[CRM\_UTM\_Source], items[Excel\_Fields], items[code], items[created\_at], items[end\_date], items[id], items[start\_date], ... | 422: detail

### **POST /api/campaigns**

- Summary: Create Campaign
- Tags: campaigns
- Request: body=CRM\_UTM\_Campaign, CRM\_UTM\_Source, code, end\_date, start\_date, status

- Responses: 201: CRM\_UTM\_Campaign, CRM\_UTM\_Source, Excel\_Fields, code, created\_at, end\_date, id, start\_date, ... | 422: detail

### **GET /api/campaigns/{campaign\_id}**

- Summary: Get Campaign  
- Tags: campaigns  
- Request: params=campaign\_id(path)  
- Responses: 200: CRM\_UTM\_Campaign, CRM\_UTM\_Source, Excel\_Fields, code, created\_at, end\_date, id, start\_date, ... | 422: detail

### **PATCH /api/campaigns/{campaign\_id}**

- Summary: Update Campaign  
- Tags: campaigns  
- Request: params=campaign\_id(path) ; body=channel, code, end\_date, name, start\_date, status  
- Responses: 200: CRM\_UTM\_Campaign, CRM\_UTM\_Source, Excel\_Fields, code, created\_at, end\_date, id, start\_date, ... | 422: detail

### **DELETE /api/campaigns/{campaign\_id}**

- Summary: Delete Campaign  
- Tags: campaigns  
- Request: params=campaign\_id(path)  
- Responses: 204: Successful Response | 422: detail

### **GET /api/campaigns/{campaign\_id}/leads**

- Summary: List Campaign Leads  
- Tags: campaigns  
- Request: params=campaign\_id(path), skip(query), limit(query)  
- Responses: 200: items[CRM\_Channel], items[CRM\_Data\_Medium], items[Excel\_Fields], items[Label\_Source\_Lead\_Status], items[campaign\_id], items[created\_at], items[current\_handler], items[current\_section], ... | 422: detail

### **GET /api/campaigns/{campaign\_id}/performance**

- Summary: Get Campaign Performance  
- Tags: campaigns  
- Request: params=campaign\_id(path)  
- Responses: 200: by\_channel, by\_metric, campaign\_id, total\_events | 422: detail

### **GET /api/customers**

- Summary: List Customers  
- Tags: customers  
- Request: params=skip(query), limit(query)  
- Responses: 200: items[CRM\_Age\_Band], items[CRM\_Education], items[CRM\_Existing\_Plan\_Flag], items[CRM\_Gender], items[CRM\_Income\_Band], items[CRM\_NonResident\_Flag], items[CRM\_Occupation], items[CRM\_Tobacco\_User], ... | 422: detail

## **POST /api/customers**

- Summary: Create Customer
- Tags: customers
- Request: body=CRM\_Age\_Band, CRM\_Education, CRM\_Existing\_Plan\_Flag, CRM\_Gender, CRM\_Income\_Band, CRM\_NonResident\_Flag, CRM\_Occupation, CRM\_Tobacco\_User, Customer\_ID
- Responses: 201: CRM\_Age\_Band, CRM\_Education, CRM\_Existing\_Plan\_Flag, CRM\_Gender, CRM\_Income\_Band, CRM\_NonResident\_Flag, CRM\_Occupation, CRM\_Tobacco\_User, ... | 422: detail

## **GET /api/customers/{customer\_id}**

- Summary: Get Customer
- Tags: customers
- Request: params=customer\_id(path)
- Responses: 200: CRM\_Age\_Band, CRM\_Education, CRM\_Existing\_Plan\_Flag, CRM\_Gender, CRM\_Income\_Band, CRM\_NonResident\_Flag, CRM\_Occupation, CRM\_Tobacco\_User, ... | 422: detail

## **PATCH /api/customers/{customer\_id}**

- Summary: Update Customer
- Tags: customers
- Request: params=customer\_id(path) ; body=CRM\_Age\_Band, CRM\_Education, CRM\_Existing\_Plan\_Flag, CRM\_Gender, CRM\_Income\_Band, CRM\_NonResident\_Flag, CRM\_Occupation, CRM\_Tobacco\_User, Customer\_ID
- Responses: 200: CRM\_Age\_Band, CRM\_Education, CRM\_Existing\_Plan\_Flag, CRM\_Gender, CRM\_Income\_Band, CRM\_NonResident\_Flag, CRM\_Occupation, CRM\_Tobacco\_User, ... | 422: detail

## **DELETE /api/customers/{customer\_id}**

- Summary: Delete Customer
- Tags: customers
- Request: params=customer\_id(path)
- Responses: 204: Successful Response | 422: detail

## **GET /api/dashboard**

- Summary: Get Dashboard
- Tags: dashboard
- Request: none
- Responses: 200: conversion\_rate, converted\_leads, open\_tasks, pending\_followups, total\_calls, total\_campaigns, total\_customers, total\_leads

## **POST /api/data/export**

- Summary: Export Data
- Tags: data
- Request: body=file\_format, resource
- Responses: 201: file\_format, file\_path, job\_id, resource, row\_count, status | 422: detail

## **GET /api/data/history**

- Summary: Get Data History

- Tags: data
- Request: params=skip(query), limit(query)
- Responses: 200: items[completed\_at], items[created\_at], items[dataset\_name], items[details], items[error\_message], items[id], items[invalid\_row\_count], items[row\_count], ... | 422: detail

## **POST /api/data/import**

- Summary: Import Data
- Tags: data
- Request: body=dataset\_name, file\_format, source\_path
- Responses: 201: completed\_at, created\_at, dataset\_name, details, error\_message, id, invalid\_row\_count, row\_count, ... | 422: detail

## **GET /api/data/import/{job\_id}**

- Summary: Get Import Job
- Tags: data
- Request: params=job\_id(path)
- Responses: 200: completed\_at, created\_at, dataset\_name, details, error\_message, id, invalid\_row\_count, row\_count, ... | 422: detail

## **GET /api/data/quality**

- Summary: Get Data Quality
- Tags: data
- Request: none
- Responses: 200: completeness, generated\_at, table\_counts

## **GET /api/data/validation**

- Summary: Get Data Validation
- Tags: data
- Request: none
- Responses: 200: generated\_at, issues, total\_issues

## **GET /api/decision/customers/{customer\_id}**

- Summary: Get Customer Decisions
- Tags: decision
- Request: params=customer\_id(path)
- Responses: 200: items[confidence], items[created\_at], items[customer\_id], items[decision\_type], items[id], items[inputs], items[lead\_id], items[priority], ... | 422: detail

## **GET /api/decision/leads/{lead\_id}**

- Summary: Get Lead Decisions
- Tags: decision
- Request: params=lead\_id(path)
- Responses: 200: items[confidence], items[created\_at], items[customer\_id], items[decision\_type], items[id], items[inputs], items[lead\_id], items[priority], ... | 422: detail

## **POST /api/decision/next-action**

- Summary: Decide Next Action
- Tags: decision
- Request: body=customer\_id, lead\_id
- Responses: 200: confidence, decision\_type, inputs, priority, rationale, recommended\_action | 422: detail

## **GET /api/engagement/email**

- Summary: List Email Events
- Tags: engagement
- Request: params=skip(query), limit(query), lead\_id(query), customer\_id(query), campaign\_id(query)
- Responses: 200: items[CRM\_Channel], items[Excel\_Fields], items[MSG\_Engaged], items[campaign\_id], items[customer\_id], items[event\_payload], items[event\_time], items[id], ... | 422: detail

## **GET /api/engagement/rcs**

- Summary: List Rcs Events
- Tags: engagement
- Request: params=skip(query), limit(query), lead\_id(query), customer\_id(query), campaign\_id(query)
- Responses: 200: items[CRM\_Channel], items[Excel\_Fields], items[MSG\_Engaged], items[campaign\_id], items[customer\_id], items[event\_payload], items[event\_time], items[id], ... | 422: detail

## **GET /api/engagement/website**

- Summary: List Website Events
- Tags: engagement
- Request: params=skip(query), limit(query), lead\_id(query), customer\_id(query), campaign\_id(query)
- Responses: 200: items[CRM\_Channel], items[Excel\_Fields], items[MSG\_Engaged], items[campaign\_id], items[customer\_id], items[event\_payload], items[event\_time], items[id], ... | 422: detail

## **GET /api/engagement/whatsapp**

- Summary: List Whatsapp Events
- Tags: engagement
- Request: params=skip(query), limit(query), lead\_id(query), customer\_id(query), campaign\_id(query)
- Responses: 200: items[CRM\_Channel], items[Excel\_Fields], items[MSG\_Engaged], items[campaign\_id], items[customer\_id], items[event\_payload], items[event\_time], items[id], ... | 422: detail

## **GET /api/followups**

- Summary: List Followups
- Tags: followups
- Request: params=skip(query), limit(query), task\_id(query), lead\_id(query), customer\_id(query)
- Responses: 200: items[Excel\_Fields], items[Label\_Basis], items[channel], items[completed\_at], items[created\_at], items[customer\_id], items[id], items[lead\_id], ... | 422: detail

## **POST /api/followups**

- Summary: Create Followup
- Tags: followups

- Request: body=Label\_Basis, channel, customer\_id, lead\_id, scheduled\_at, status, task\_id
- Responses: 201: Excel\_Fields, Label\_Basis, channel, completed\_at, created\_at, customer\_id, id, lead\_id, ... | 422: detail

## **PATCH /api/followups/{followup\_id}**

- Summary: Update Followup
- Tags: followups
- Request: params=followup\_id(path) ; body=channel, completed\_at, customer\_id, lead\_id, notes, scheduled\_at, status, task\_id
- Responses: 200: Excel\_Fields, Label\_Basis, channel, completed\_at, created\_at, customer\_id, id, lead\_id, ... | 422: detail

## **GET /api/health**

- Summary: Health Check
- Tags: health
- Request: none
- Responses: 200: database, status

## **GET /api/journey/leads/{lead\_id}**

- Summary: Get Lead Journey Summary
- Tags: journey
- Request: params=lead\_id(path)
- Responses: 200: last\_event\_time, lead\_id, total\_events, unique\_event\_names | 422: detail

## **GET /api/journey/website**

- Summary: List Website Events
- Tags: journey
- Request: params=skip(query), limit(query), lead\_id(query), customer\_id(query), device\_type(query)
- Responses: 200: items[Excel\_Fields], items[WEB\_Device\_Type], items[WEB\_New\_Vs\_Repeat], items[WEB\_Step\_Name], items[WEB\_Step\_Number], items[customer\_id], items[event\_name], items[event\_payload], ... | 422: detail

## **GET /api/leads**

- Summary: List Leads
- Tags: leads
- Request: params=skip(query), limit(query), customer\_id(query), campaign\_id(query)
- Responses: 200: items[CRM\_Channel], items[CRM\_Data\_Medium], items[Excel\_Fields], items[Label\_Source\_Lead\_Status], items[campaign\_id], items[created\_at], items[current\_handler], items[current\_section], ... | 422: detail

## **POST /api/leads**

- Summary: Create Lead
- Tags: leads
- Request: body=CRM\_Channel, CRM\_Data\_Medium, Label\_Source\_Lead\_Status, campaign\_id, current\_handler, current\_section, current\_stage, customer\_id, excel\_fields, lead\_score, ...

- Responses: 201: CRM\_Channel, CRM\_Data\_Medium, Excel\_Fields, Label\_Source\_Lead\_Status, campaign\_id, created\_at, current\_handler, current\_section, ... | 422: detail

### **GET /api/leads/{lead\_id}**

- Summary: Get Lead  
- Tags: leads  
- Request: params=lead\_id(path)  
- Responses: 200: CRM\_Channel, CRM\_Data\_Medium, Excel\_Fields, Label\_Source\_Lead\_Status, campaign\_id, created\_at, current\_handler, current\_section, ... | 422: detail

### **PATCH /api/leads/{lead\_id}**

- Summary: Update Lead  
- Tags: leads  
- Request: params=lead\_id(path) ; body=campaign\_id, current\_handler, current\_section, current\_stage, customer\_id, excel\_fields, lead\_score, priority, recommended\_action, source\_channel, ...  
- Responses: 200: CRM\_Channel, CRM\_Data\_Medium, Excel\_Fields, Label\_Source\_Lead\_Status, campaign\_id, created\_at, current\_handler, current\_section, ... | 422: detail

### **DELETE /api/leads/{lead\_id}**

- Summary: Delete Lead  
- Tags: leads  
- Request: params=lead\_id(path)  
- Responses: 204: Successful Response | 422: detail

### **POST /api/leads/{lead\_id}/assign**

- Summary: Assign Lead  
- Tags: leads  
- Request: params=lead\_id(path) ; body=details, reason, related\_decision\_id, to\_handler, to\_section, trigger  
- Responses: 200: Excel\_Fields, assignment\_type, created\_at, customer\_id, details, from\_handler, from\_section, id, ... | 422: detail

### **GET /api/leads/{lead\_id}/calls**

- Summary: Get Lead Calls  
- Tags: leads  
- Request: params=lead\_id(path), skip(query), limit(query)  
- Responses: 200: items[CDR\_Avg\_Talk\_Sec], items[CDR\_Call\_Direction], items[CDR\_Top\_Call\_Status], items[Excel\_Fields], items[created\_at], items[customer\_id], items[ended\_at], items[id], ... | 422: detail

### **GET /api/leads/{lead\_id}/details**

- Summary: Get Lead Details  
- Tags: leads  
- Request: params=lead\_id(path)  
- Responses: 200: ai, assignment, calls, campaign, customer, engagement, followups, journey, ... | 422: detail

### **GET /api/leads/{lead\_id}/journey**



- Summary: Get Lead Journey
- Tags: leads
- Request: params=lead\_id(path), skip(query), limit(query)
- Responses: 200: items[Excel\_Fields], items[WEB\_Device\_Type], items[WEB\_New\_Vs\_Repeat], items[WEB\_Step\_Name], items[WEB\_Step\_Number], items[customer\_id], items[event\_name], items[event\_payload], ... | 422: detail

## **GET /api/leads/{lead\_id}/outcomes**

- Summary: List Lead Outcomes
- Tags: leads
- Request: params=lead\_id(path)
- Responses: 200: items[Excel\_Fields], items[action\_type], items[created\_at], items[customer\_id], items[details], items[followup\_required], items[id], items[lead\_id], ... | 422: detail

## **POST /api/leads/{lead\_id}/outcomes**

- Summary: Record Lead Outcome
- Tags: leads
- Request: params=lead\_id(path) ; body=action\_type, details, followup\_required, next\_action\_hint, notes, outcome\_code, outcome\_label
- Responses: 201: Excel\_Fields, action\_type, created\_at, customer\_id, details, followup\_required, id, lead\_id, ... | 422: detail

## **GET /api/leads/{lead\_id}/timeline**

- Summary: Get Lead Timeline
- Tags: leads
- Request: params=lead\_id(path)
- Responses: 200: items[Excel\_Fields], items[created\_at], items[customer\_id], items[details], items[event\_source], items[event\_type], items[id], items[lead\_id] | 422: detail

## **POST /api/leads/{lead\_id}/transfer**

- Summary: Transfer Lead
- Tags: leads
- Request: params=lead\_id(path) ; body=details, reason, related\_decision\_id, to\_handler, to\_section, trigger
- Responses: 200: Excel\_Fields, assignment\_type, created\_at, customer\_id, details, from\_handler, from\_section, id, ... | 422: detail

## **GET /api/ml/models**

- Summary: List Models
- Tags: ml
- Request: params=skip(query), limit(query)
- Responses: 200: items[created\_at], items[latest\_job\_id], items[model\_name], items[status], items[training\_rows] | 422: detail

## **POST /api/ml/train**

- Summary: Train Model

- Tags: ml
- Request: body=model\_name
- Responses: 201: artifact\_path, completed\_at, created\_at, error\_message, id, metrics, model\_name, status, ... | 422: detail

### **GET /api/ml/train/{job\_id}**

- Summary: Get Training Job
- Tags: ml
- Request: params=job\_id(path)
- Responses: 200: artifact\_path, completed\_at, created\_at, error\_message, id, metrics, model\_name, status, ... | 422: detail

### **GET /api/products**

- Summary: List Products
- Tags: products
- Request: params=skip(query), limit(query)
- Responses: 200: items[CRM\_Product\_Code], items[CRM\_Product\_Name], items[Excel\_Fields], items[created\_at], items[id], items[updated\_at] | 422: detail

### **POST /api/products**

- Summary: Create Product
- Tags: products
- Request: body=CRM\_Product\_Code, CRM\_Product\_Name
- Responses: 201: CRM\_Product\_Code, CRM\_Product\_Name, Excel\_Fields, created\_at, id, updated\_at | 422: detail

### **GET /api/products/{product\_id}**

- Summary: Get Product
- Tags: products
- Request: params=product\_id(path)
- Responses: 200: CRM\_Product\_Code, CRM\_Product\_Name, Excel\_Fields, created\_at, id, updated\_at | 422: detail

### **PATCH /api/products/{product\_id}**

- Summary: Update Product
- Tags: products
- Request: params=product\_id(path) ; body=code, name
- Responses: 200: CRM\_Product\_Code, CRM\_Product\_Name, Excel\_Fields, created\_at, id, updated\_at | 422: detail

### **DELETE /api/products/{product\_id}**

- Summary: Delete Product
- Tags: products
- Request: params=product\_id(path)
- Responses: 204: Successful Response | 422: detail

### **GET /api/reports/campaign-performance**

- Summary: Get Campaign Performance
- Tags: reports
- Request: none
- Responses: 200: items

## **GET /api/reports/pipeline**

- Summary: Get Pipeline
- Tags: reports
- Request: none
- Responses: 200: converted\_leads, lost\_leads, new\_leads, qualified\_leads, total\_leads

## **GET /api/reports/workload**

- Summary: Get Workload
- Tags: reports
- Request: none
- Responses: 200: completed\_followups, completed\_tasks, in\_progress\_tasks, open\_tasks, pending\_followups, total\_followups, total\_tasks

## **GET /api/tasks**

- Summary: List Tasks
- Tags: tasks
- Request: params=skip(query), limit(query), lead\_id(query), customer\_id(query), status\_filter(query)
- Responses: 200: items[Excel\_Fields], items[Label\_Source\_Disposition], items[created\_at], items[customer\_id], items[description], items[due\_at], items[id], items[lead\_id], ... | 422: detail

## **POST /api/tasks**

- Summary: Create Task
- Tags: tasks
- Request: body=Label\_Source\_Disposition, customer\_id, description, due\_at, lead\_id, priority, status
- Responses: 201: Excel\_Fields, Label\_Source\_Disposition, created\_at, customer\_id, description, due\_at, id, lead\_id, ... | 422: detail

## **GET /api/tasks/{task\_id}**

- Summary: Get Task
- Tags: tasks
- Request: params=task\_id(path)
- Responses: 200: Excel\_Fields, Label\_Source\_Disposition, created\_at, customer\_id, description, due\_at, id, lead\_id, ... | 422: detail

## **PATCH /api/tasks/{task\_id}**

- Summary: Update Task
- Tags: tasks
- Request: params=task\_id(path) ; body=customer\_id, description, due\_at, lead\_id, priority, status, title

- Responses: 200: Excel\_Fields, Label\_Source\_Disposition, created\_at, customer\_id, description, due\_at, id, lead\_id, ... | 422: detail

## **DELETE /api/tasks/{task\_id}**

- Summary: Delete Task
- Tags: tasks
- Request: params=task\_id(path)
- Responses: 204: Successful Response | 422: detail

## **GET /health**

- Summary: Health Check
- Tags: health
- Request: none
- Responses: 200: database, status

## **8. Lead Lifecycle Behavior**

1. New lead creation writes lead\_created event to timeline.
2. Assign/transfer updates lead ownership and section, and writes lifecycle events.
3. Outcome recording can update lead status/stage and auto-create follow-ups.
4. /leads/{id}/details returns a unified payload for frontend lead workspace rendering.

## **9. Validation Snapshot**

- Test suite passing (including lifecycle tests).
- Live smoke APIs passing with real sample data preloaded.
- Lifecycle endpoints verified: timeline, details, assign, transfer, outcomes list/create.